

Math 871 Topology I Worksheet 4

1. Let $X = [0, 1] \times [0, 1] \subseteq \mathbb{R}^2$. Let $\mathcal{T}_{\text{sub}}$ be the subspace topology X inherits from $\mathbb{R}^2$, and let $\mathcal{T}_{<}$ be the dictionary order topology on X . Show that $\mathcal{T}_{\text{sub}}$ and $\mathcal{T}_{<}$ are incomparable topologies.

Note: to avoid confusion between points and intervals, let us write

$$]A, B[:= \{x \in X \mid A < x < B\}$$

for the open interval from A to B .

2. Let $Y \subseteq X$ and let $\mathcal{T}$ and $\mathcal{T}'$ be topologies on X such that $\mathcal{T}'$ is strictly finer than $\mathcal{T}$. Let $\mathcal{T}_Y$ and $\mathcal{T}'_Y$ be the corresponding subspace topologies on Y .

- (a) Prove that $\mathcal{T}'_Y$ is finer than $\mathcal{T}_Y$.
- (b) Give an example showing that $\mathcal{T}'_Y$ might not be strictly finer than $\mathcal{T}_Y$.

3. Consider $\mathbb{R}$ under the included point topology with included point 0.

- (a) Find $\overline{\{0\}}$.
- (b) Find $\overline{\{1\}}$.

4. Let Y be a subspace of X and S is any subset of Y . Show that

$$\overline{S}^Y = \overline{S}^X \cap Y.$$

5. We say a topological space is T_0 if for every pair of distinct points x and y , there exists an open neighborhood U of x such that $y \notin U$ or there exists an open neighborhood V of y such that $x \notin V$. Note that if X is T_1 , then it is also T_0 . Given an example of a T_0 space that is not T_1 .
6. A topological space X has the $T_{2\frac{1}{2}}$ property if for every pair of distinct points a and b , there are open sets U and V such that $a \in U$, $b \in V$, and $\overline{U} \cap \overline{V} = \emptyset$. Show that if S is a subspace of a $T_{2\frac{1}{2}}$ space X , then S is also $T_{2\frac{1}{2}}$.